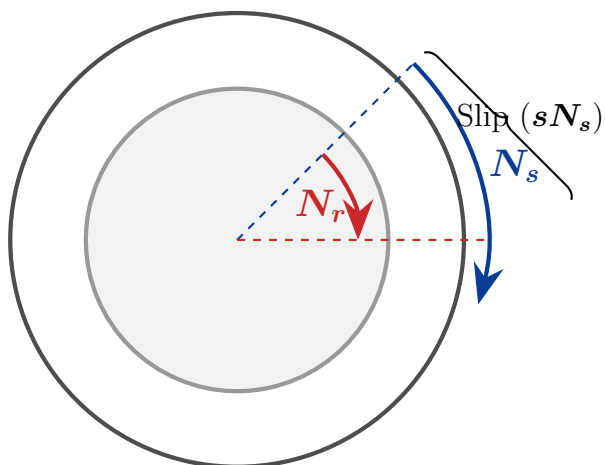


PHASE 2: KINEMATICS

& The Generalized Transformer

The Physics of the Chase

Why the rotor spins, why it can never catch up, and why an induction motor is just a transformer that learned how to run.



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Block 1: The Chase Begins

In Phase 1, we successfully created a Rotating Magnetic Field (RMF) that spins around the stator at Synchronous Speed (N_s).

Now, imagine we place a solid steel cylinder (the rotor) wrapped in closed loops of copper wire inside this spinning tornado of magnetism. What happens at the exact millisecond we turn the power on?

The Mental Video (Play this in your head!)

Frame 1 (Time = 0.000s): The RMF instantly snaps into existence and begins spinning at 1500 RPM (N_s). The heavy iron rotor is sitting perfectly still at 0 RPM ($N_r = 0$).

Because the rotor is stationary and the magnetic field is flying past it at 1500 RPM, the invisible magnetic flux lines are violently smashing through the copper bars of the rotor.

Frame 2 (Time = 0.001s): Faraday's Law awakens. Because flux lines are cutting the copper bars, a massive voltage is induced in the rotor. Because the rotor bars are short-circuited (connected together at the ends), a massive current begins to flow.

Frame 3 (Time = 0.002s): Lenz's Law awakens. The universe hates change. The rotor currents create their own magnetic field to fight the flux cutting. How does the rotor stop the RMF from flying past it? *By running after it.* The rotor develops electromagnetic torque and begins to accelerate in the exact same direction as the RMF. The chase has begun!

Block 2: The Concept of Slip (s)

The rotor accelerates. 100 RPM... 500 RPM... 1000 RPM. As it speeds up, it is getting closer to the speed of the RMF (1500 RPM).

The Great Paradox

Can the rotor ever catch the RMF? Can $N_r = N_s$?

Let's pretend it does. Imagine the rotor is now spinning at exactly 1500 RPM, right alongside the RMF.

- If both the rotor and the magnetic field are moving at the exact same speed, the relative speed between them is **Zero**.
- If the relative speed is zero, the magnetic flux lines are no longer "cutting" the copper bars. They are just moving together.
- By Faraday's Law, if there is no flux cutting ($\frac{d\Phi}{dt} = 0$), there is NO induced voltage.
- No voltage = No current.
- No current = No magnetic force. No torque.

Without torque, friction and wind resistance instantly slow the rotor down.

The rotor is cursed. It spins *because* it is being cut by the flux. If it catches the flux, the cutting stops, and it loses its power. Therefore, an induction motor can **never** run at

synchronous speed. It must always "slip" slightly behind the RMF to ensure flux lines keep cutting its bars.

Mathematical Definition of Slip

The difference between the Synchronous Speed (N_s) and the actual Rotor Speed (N_r) is called the **Slip Speed**.

$$\text{Slip Speed} = N_s - N_r \quad (1)$$

We usually express this as a fraction or percentage of the synchronous speed, simply called **Slip** (s):

$$s = \frac{N_s - N_r}{N_s} \quad (2)$$

From this, we can easily find the rotor's actual speed:

$$N_r = N_s(1 - s) \quad (3)$$

Typical Values: At starting, $N_r = 0$, so $s = 1$ (or 100%). Under normal full-load running conditions, the motor only needs to slip a tiny bit to generate enough torque, so s is usually between 0.02 and 0.05 (2% to 5%).

Block 3: The Kinematics of Magnetic Fields (Exam Mastery)

Semester Exam Trap

This specific block contains the answers to the most frequently asked, deeply confusing conceptual questions in your semester exams (e.g., 2023 Q1a, 2024 Q1c). Examiners will try to trick you with the phrase "relative speed of X with respect to Y." You must map this out neurologically.

Let's establish our three entities and their absolute physical speeds in space (relative to the stationary ground/stator):

1. **The Stator Frame:** Absolute speed = 0.
2. **The Stator RMF:** Absolute speed = N_s .
3. **The Rotor Body:** Absolute speed = $N_r = N_s(1 - s)$.

Now, let's analyze the **Rotor's Magnetic Field**. Because the stator field cuts the rotor at a relative speed of $(N_s - N_r) = sN_s$, it induces alternating currents in the rotor. The frequency of these rotor currents is the **Slip Frequency**:

$$f_2 = sf_1 \quad (4)$$

where f_1 is the 50Hz grid frequency.

These alternating rotor currents flow through the 3-phase rotor windings and create *their own* Rotating Magnetic Field. How fast does this Rotor Field spin? The speed of any magnetic

field is given by $120f/P$.

$$N_{rotor_field} = \frac{120f_2}{P} = \frac{120(sf_1)}{P} = s \left(\frac{120f_1}{P} \right) = sN_s \quad (5)$$

CRITICAL DISTINCTION: This speed (sN_s) is the speed of the rotor's magnetic field *relative to the rotor body itself*. The rotor body is already physically spinning at N_r .

What is the absolute speed of the Rotor's Magnetic Field in space?

$$\begin{aligned} \text{Space Speed of Rotor Field} &= \text{Mechanical Rotor Speed} + \text{Relative Field Speed} \\ &= N_r + sN_s \\ &= N_s(1 - s) + sN_s \\ &= N_s - sN_s + sN_s \\ &= N_s \end{aligned} \quad (6)$$

The Kinematic Cheat Sheet

Memorize these five relative speeds. They are the keys to acing Part I short questions.

- Speed of Stator RMF with respect to Stator = N_s
- Speed of Rotor with respect to Stator = N_r
- Speed of Stator RMF with respect to Rotor = sN_s
- Speed of Rotor RMF with respect to Rotor = sN_s
- Speed of Rotor RMF with respect to Stator = N_s

The Ultimate Conclusion: Since the Stator RMF spins at N_s in space, and the Rotor RMF *also* spins at N_s in space, **the relative speed between the Stator Magnetic Field and the Rotor Magnetic Field is exactly ZERO**. They are magnetically locked together, spinning hand-in-hand to produce steady torque!

Block 4: The "Generalized Transformer" Concept

In the 2022 and 2024 exams, you were asked: "*Why is an induction machine called a generalized transformer?*" We finally have the physics to prove it.

A standard electrical transformer has a primary winding and a secondary winding on a shared iron core.

- You apply 50Hz AC to the primary.
- It creates an alternating magnetic flux.
- This flux cuts the secondary.
- It induces a 50Hz AC voltage in the secondary.
- The secondary does not physically move.

Look at our Induction Motor.

- We apply 50Hz AC to the Stator (Primary).
- It creates a rotating magnetic flux.

- This flux cuts the Rotor (Secondary) across an air gap.
- It induces a voltage in the Rotor.

The Mental Video (Play this in your head!)

Here is the magic difference. In a transformer, the frequency of the secondary is **always** equal to the primary ($f_2 = f_1$).

In an induction motor, the rotor is allowed to spin. As it spins, it chases the magnetic field, which reduces the relative speed of the flux cutting it. Because the flux is cutting it slower, the frequency of the induced voltage drops!

The Mathematical Link

Let E_2 be the voltage induced in the rotor when it is physically blocked from spinning (Standstill, $N_r = 0$). At standstill, slip $s = \frac{N_s - 0}{N_s} = 1$.

When $s = 1$:

- The relative speed of flux cutting is 100% (N_s).
- The rotor voltage is at its maximum: E_2 .
- The rotor frequency is full grid frequency: $f_2 = 1 \times f_1 = 50\text{Hz}$.

At $s = 1$, the induction motor is literally just a short-circuited transformer with an air gap.

Now, let the motor spin normally at slip s :

- The relative speed of flux cutting drops to a fraction: $s \times N_s$.
- Because Faraday's Law says voltage is proportional to the rate of flux cutting, the induced voltage scales down perfectly with slip: $E_{2s} = sE_2$.
- The frequency scales down perfectly with slip: $f_2 = sf_1$.
- The leakage reactance (which is $2\pi fL$) also scales down with slip: $X_{2s} = sX_2$.

The Final Answer for the Exam

An induction motor is a **Generalized Transformer** because the static transformer is merely a restricted, special case of the induction machine.

The induction machine equation ($f_2 = sf_1$) governs all electromagnetic induction across two circuits. A standard transformer is simply an induction machine where the slip s is permanently forced to be 1 because the secondary is physically bolted down and cannot rotate. By allowing the secondary to rotate, the induction machine generalizes the concept of transformer action to include variable frequency and mechanical power conversion.

Phase 2 Completion Checklist

You have successfully mapped the physical motion of the motor to its mathematical slip parameters.

Ensure you can answer "Yes" to these checkpoints before moving to Phase 3:

- [] Do I understand why the rotor can NEVER reach synchronous speed?
- [] Can I confidently state that the relative speed between the Stator's Magnetic Field and the Rotor's Magnetic Field is ZERO?
- [] Do I know why rotor frequency (f_2) drops as the motor speeds up?
- [] Can I explain to an examiner why an induction motor is a transformer that is allowed to spin?

The Mental Video (Play this in your head!)

The Bridge to Phase 3: We have mastered the pure physics. We know how the fields spin and how the rotor chases them. But so far, we have imagined perfectly smooth, ideal sine waves and ideal magnetic fields.

In the real world, the stator is not a perfectly smooth ring. It has teeth and slots. The coils are not perfect circles; they are spread out (distributed) and sometimes shortened (short-pitched).

Because of these real-world manufacturing realities, our perfect Magnetic Field gets distorted. It creates "Harmonics" — parasitic mini-magnetic fields that try to make the motor spin backwards or get stuck at low speeds. In Phase 3, we will descend into the dirty, imperfect reality of machine construction and conquer "Cogging", "Crawling", and "Space Harmonics."